

EDUCATION

Master of Applied Science in Biomedical Engineering <i>University of Ottawa</i> - Thesis: Design and Evaluation of a Mobile App for Computer Vision–Based Clinical Assessment of Movement - Supervisors: Dr. Natalie Baddour & Dr. Edward Lemaire	Ottawa, Ontario 2026
Honours Bachelor of Science in Computer Science, Cum Laude <i>University of Ottawa</i>	Ottawa, Ontario 2024

RESEARCH EXPERIENCE

Graduate Researcher <i>Movealytics Research Lab, University of Ottawa & CHEO</i> - Standardized clinical testing protocols for CHEO researchers by developing a secure mobile app that guides non-expert users through data acquisition. - Automated data quality validation algorithms, including real-time pose detection, multiple object identification, and pitch detection to provide immediate user feedback, ensuring the input video met the requirements for accurate clinical grading. - Optimized clinical data workflows by implementing background video processing and asynchronous multi-video uploads, reducing user-perceived latency while maintaining secure, high-integrity data transfer using Azure.	September 2024 – Present Ottawa, Ontario
Undergraduate Researcher <i>Movealytics Research Lab, University of Ottawa & CHEO</i> - Designed Flutter prototype and data workflows for video-based clinical motion assessments. - Worked with researchers to translate requirements into functional prototypes.	January 2024 – April 2024 Ottawa, Ontario

PROFESSIONAL EXPERIENCE

Database Support Analyst <i>Centre d'Enrichissement Sportif et Académique de Montréal</i> - Managed SQL database system for 500+ students and 40 staff, ensuring accurate records - Built queries and reports to support operations and maintain data reliability	January 2023 – April 2024 Montréal, Quebec (On-site)
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AI/ML RESEARCH PROJECTS

mRNA Localization Prediction <i>Python, CatBoost, XGBoost, LightGBM, Bioinformatics</i> - Achieved a peak test accuracy of 82.1% and a macro precision of 90.9% for multi-class subcellular localization by engineering a high-dimensional k-mer feature matrix (k=1-5) and optimizing LightGBM, XGBoost, and CatBoost models. - Validated sequence-based localization signals by developing a reproducible Python/Jupyter workflow and conducting feature importance analysis to identify critical genomic motifs such as "AAAAA" and "CCT".	2025
Melanoma Detection with CNNs <i>Python, TensorFlow, Computer Vision, Medical Imaging</i> - Boosted screening validation accuracy from 60% to 91% by designing a custom Convolutional Neural Network (CNN) in TensorFlow and implementing class-balancing synthetic data augmentation on the 2,300+ image ISIC dataset. - Architected and deployed an end-to-end medical diagnostic platform featuring a React frontend and a Flask-based REST API on Heroku to provide low-latency clinical inference.	2025
Fatigue Detection from EMG Signals <i>Python, Arduino, MyoWare</i> - Identified muscle fatigue thresholds in dynamic exercise protocols by processing 220,000+ EMG data points using 4th-order Butterworth bandpass filtering and rolling average smoothing in Python. - Quantified physiological fatigue progression through the extraction of time and frequency domain features (RMS, MAV, MDF), correlating spectral shifts with motor unit recruitment patterns.	2025

TECHNICAL SKILLS

Languages & Frameworks: Python, SQL, Flutter, TensorFlow, PyTorch, Scikit-learn. ML Methods: Gradient Boosting (XGBoost/LightGBM/CatBoost), CNNs, Sequence Analysis, Feature Engineering. DevOps: Azure, Docker, Git, CI/CD.
